

Uniform and Automated Deployment in SCVU

Bootc for Air-Gapped, Reproducible Builds

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1. Problem

How do you ensure uniform software deployment across machines in an air-gapped environment without spending hours configuring each one?

2. Solution

Bootc (Bootable Containers)

Bootc combines OS + Applications + Configs into **1** image that can be **atomically deployed** to any workstation.

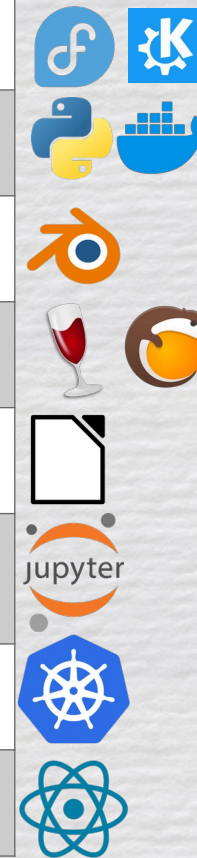
3. Key Features of Bootc

- Immutable Root Filesystem
- Container-based builds (OCI)
- Rollback and Updates
- Fast and Reproducible Deployments

4. Workflow



Stack Breakdown	
Stack(s)	Application(s)
Core OS & Desktop	Fedora 43 & KDE
Development	Python, Git, Docker
Graphics & Design	CUDA, Blender
Windows Emulation	Wine, Lutris
Office	LibreOffice
Machine Learning	Numpy, pandas, Jupyter
Deployment	Openshift, Kubernetes
JS/Frontend	React, Vue, Typescript



5. Value Proposition

- Reproducible
- Offline-capable
- Fast Deployment
- Customizable
- Reduced support

6. Limitations

- Large ISO file size (15GB)
- Certain hardware may have issues.
- Immutable Root = Rebuild for changes
- Offline packages become stale
- Linux/Container learning curve

7. Next Steps

- Pilot with incoming trainees
- Image Variants
- Repository upgrades
- Prep for unit wide deployment

8. Conclusion

Using Bootc allows us to deploy systems quicker, more consistently and more efficiently.